

Community Composting 101



Composting
Fundamentals
& Recipe
Development

Neighborhood
Soil Rebuilders
COMPOSTER TRAINING PROGRAM

ILSR
INSTITUTE FOR
Local Self-Reliance

Composting Fundamentals & Recipe Development

- ✓ The Compost Pile Food Web
- ✓ Types of Composting
- ✓ Important Ingredients
- ✓ Developing a Composting Recipe
- ✓ Temperature & The Stages of Composting

Compost = Magic?



Photos: ILSR



ILSR Recreates D.L. Dindal's original illustration from 1978.

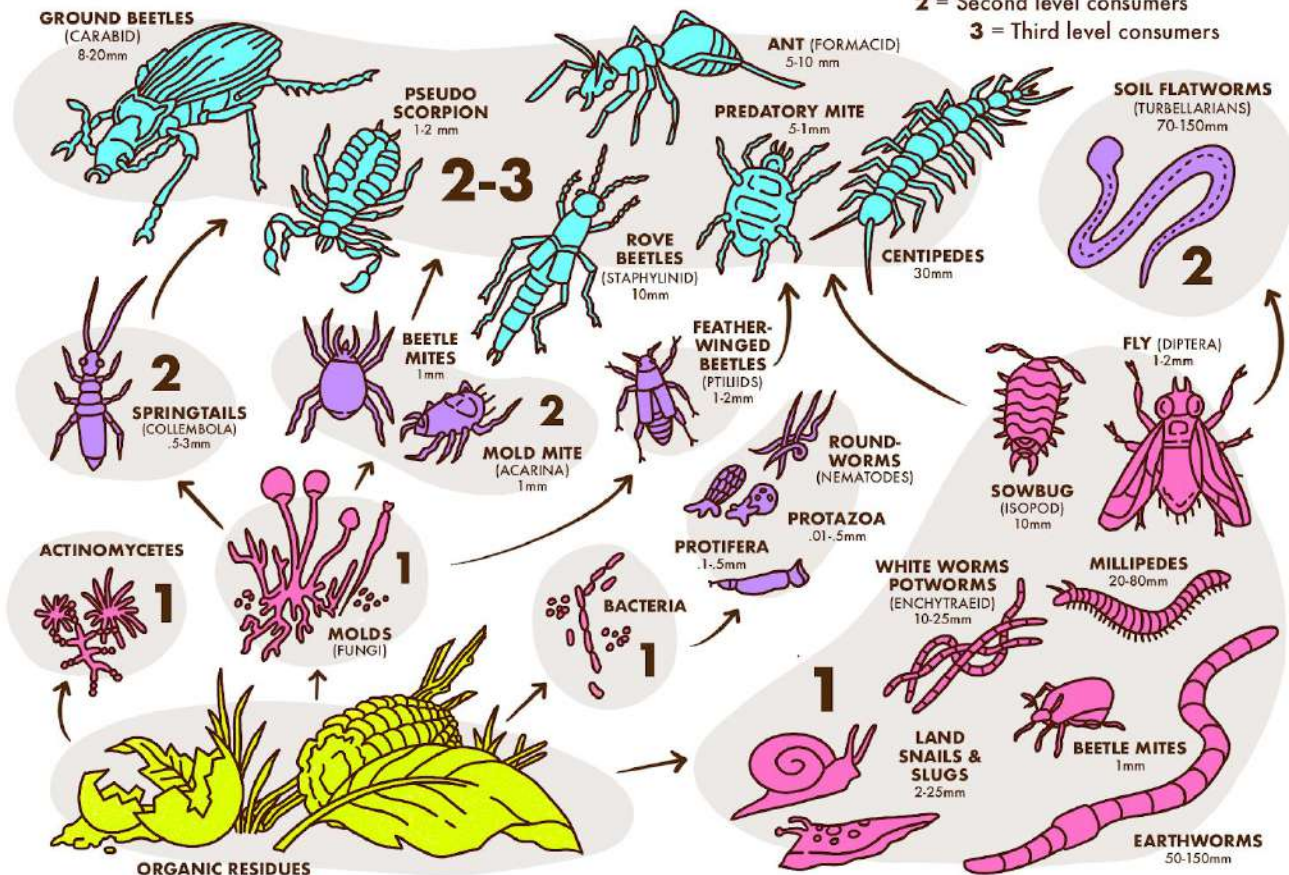
THE COMPOST PILE FOOD WEB

The energy flows in the direction of the arrow.

1 = First level consumers

2 = Second level consumers

3 = Third level consumers



Cold Composting vs. Hot Composting

Passive or Cold

- Low-effort
- Little attention to watering or optimizing conditions
- Slower (take a year)
- Weed seeds & pathogens may persist

Active or Hot

- More attention
- Pile regularly turned
- Adequate volume reached (~1 cubic yard)
- Adequate moisture maintained
- Quicker (6 months)
- Pathogens reduced
- Weed seeds killed



Cold or Passive Composting

- Low-effort
- Little attention to watering or optimizing conditions
- Slower (takes a year)
- Weed seeds & pathogens may persist



Photo: @RustBeltRiders @Tilth.Soil



Vermicomposting



Source: ILSR



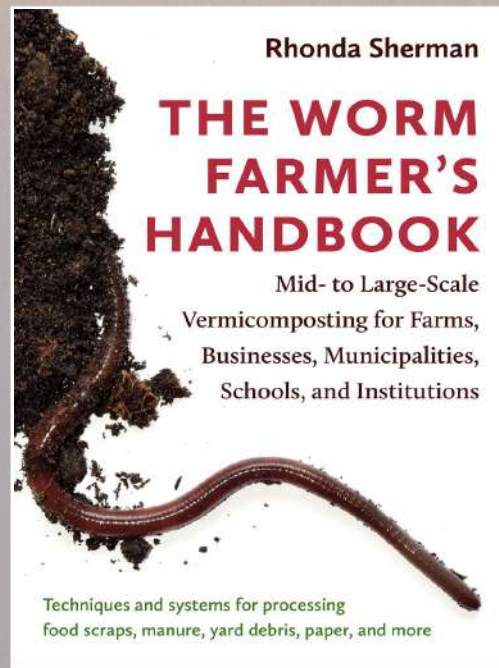
Source: Chris Cano



Vermicomposting



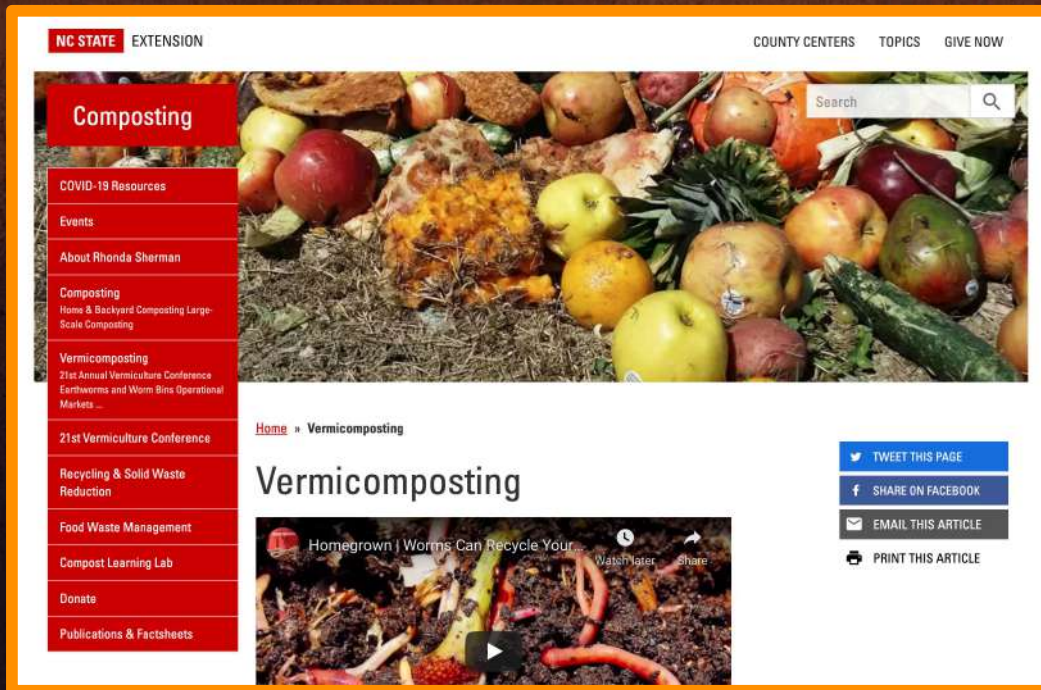
Webinar Resources: Composting with Worms on a Mid to Large-Scale – What, Why, How, and Who



<https://ilsr.org/composting/>



Vermicomposting



<https://composting.ces.ncsu.edu/vermicomposting-2/>



Active or Hot Composting

- More attention
- Higher temperatures
- Pile regularly turned
- Adequate moisture maintained
- Adequate volume reached (~1 cubic yard)
- Quicker (6 months)
- Pathogens reduced
- Weed seeds killed

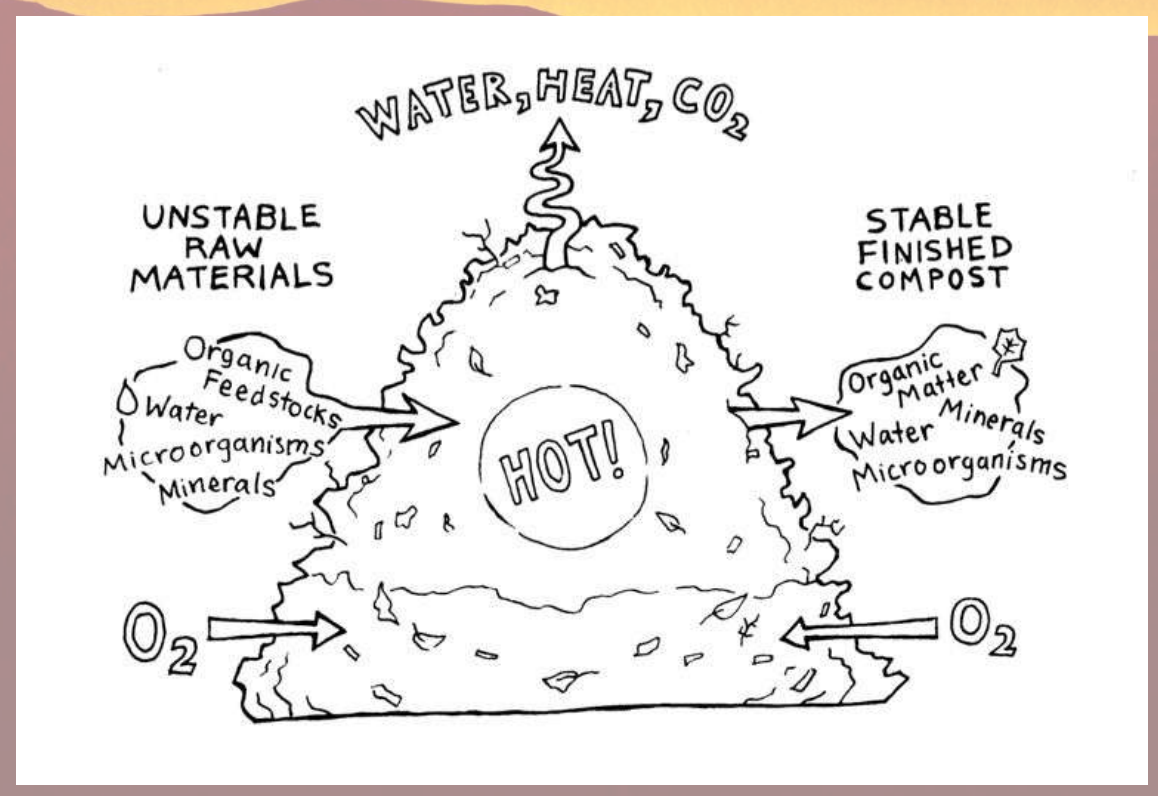


Source: ILSR

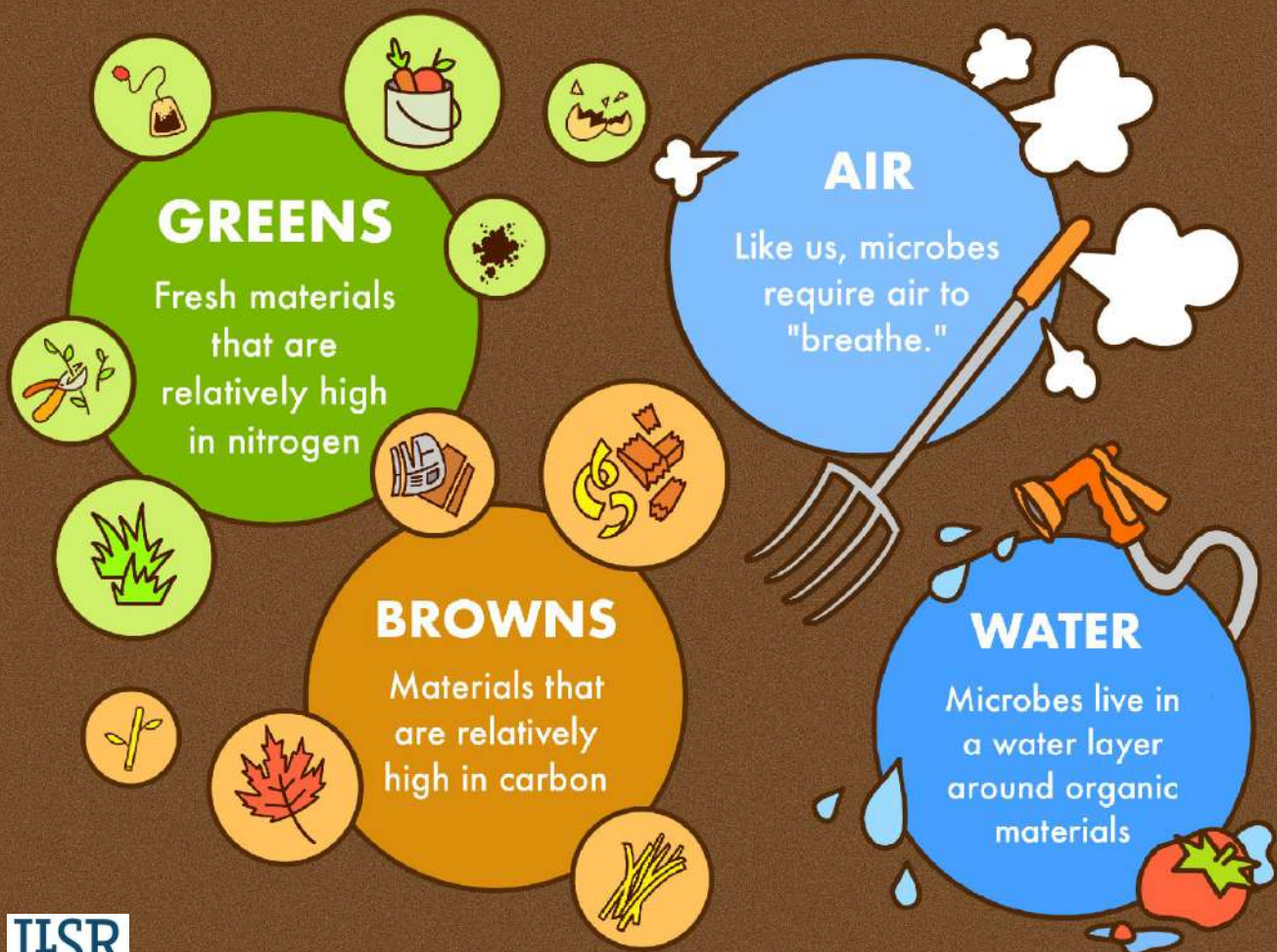


BK ROT, Bushwick, NYC.
Photo: Valerie Rizzo.

The Hot Composting Process

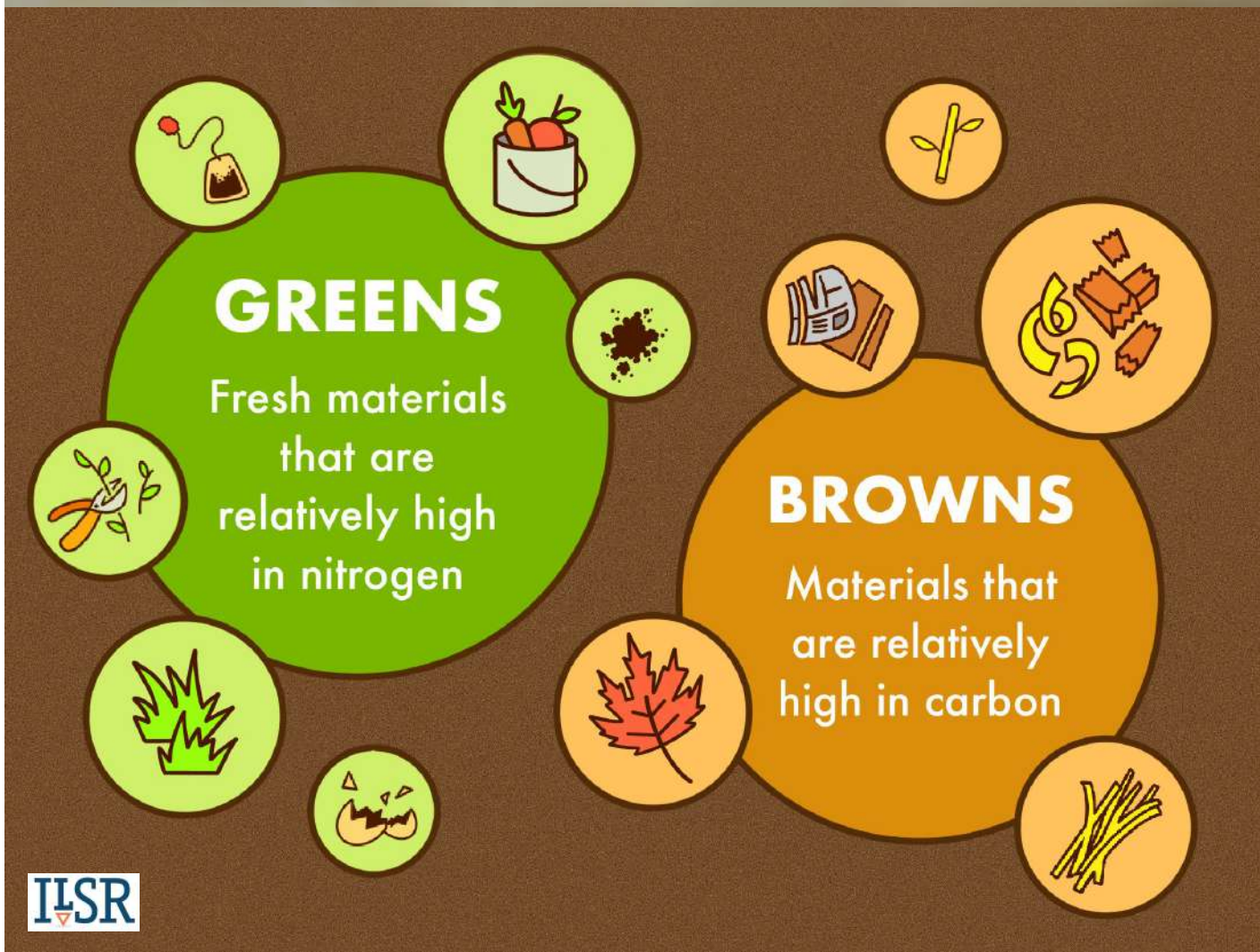


Source: Adapted from Benny Erez, "The Composting Process," Neighborhood Soil Rebuilders Composter Training Program.





Source: "The Great Pandemic Bake-Off May Be Over," NPR News, 2020.
Image: Karin Dreyer / Getty Images / Tetra Images RF



GREENS

Fresh materials
that are
relatively high
in nitrogen



BROWNS

Materials that
are relatively
high in carbon



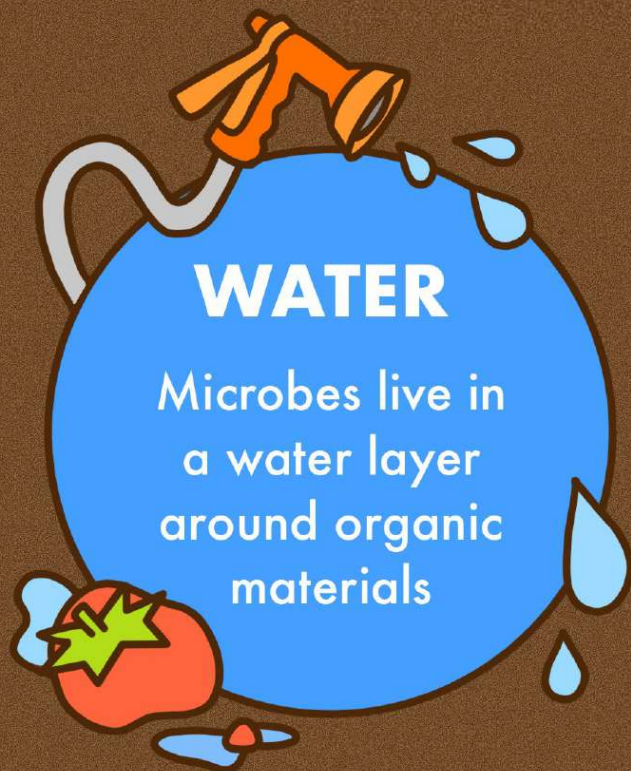
WATER

Microbes live in
a water layer
around organic
materials



AIR

Like us, microbes
require air to
"breathe."



WATER

Microbes live in a water layer around organic materials

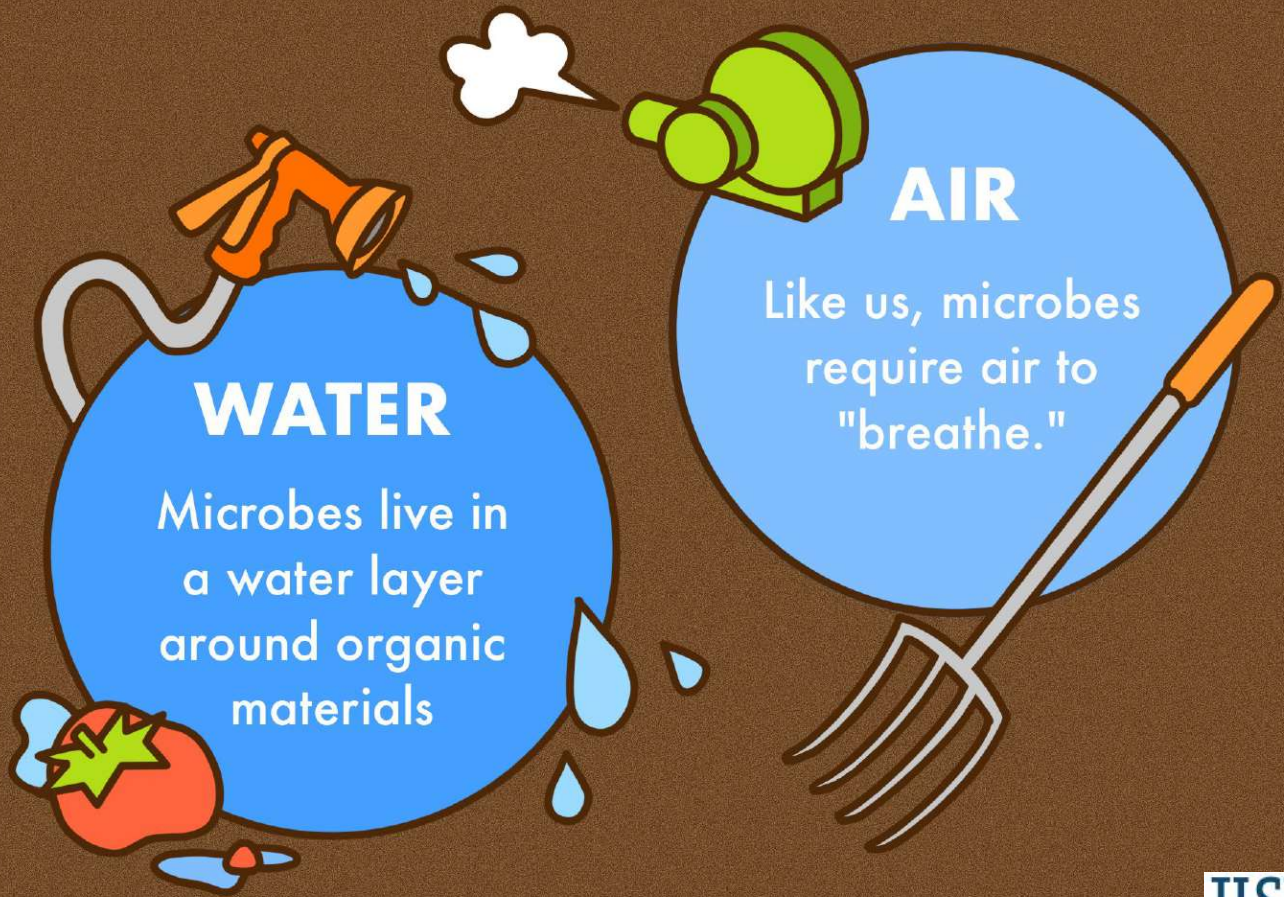


Source: ILSR



AIR

Like us, microbes require air to "breathe."



Your Composting Recipe



The Carbon to Nitrogen Ratio (C:N)

C:N ratio = Weight of C ÷ Weight of N
(on a *dry weight* basis)

Ideal C:N Ratio = 25 - 30:1
(At the beginning of the composting process)



Carbon to Nitrogen Ratio (C:N)

Recipe calculator

Feedstocks

Water 15 Gallons

Food Waste WSU
N = 2.5% C = 37.1% C:N = 14.84 1 cy
BD = 1517 lb/cy Moisture = 69%

Straw, Stable Bedding
N = 0.4% C = 50.8% C:N = 127 .5 cy
BD = 227 lb/cy Moisture = 15%

Leaves
N = 0.9% C = 48.6% C:N = 54 1.5 cy
BD = 455.1 lb/cy Moisture = 45%

Recipe Results 3 feedstocks

	Results	Recommended
C:N	26:1	20:1 - 40:1
Moisture	61.4%	40 - 65%
Bulk density	813 lb/cy	800 - 1,000 lb/cy

Online Calculator:

<https://app.compostcalc.com/calculator>

Field Guides:

NRAES *On-Farm Composting Handbook & Field Guide to On-Farm Composting*

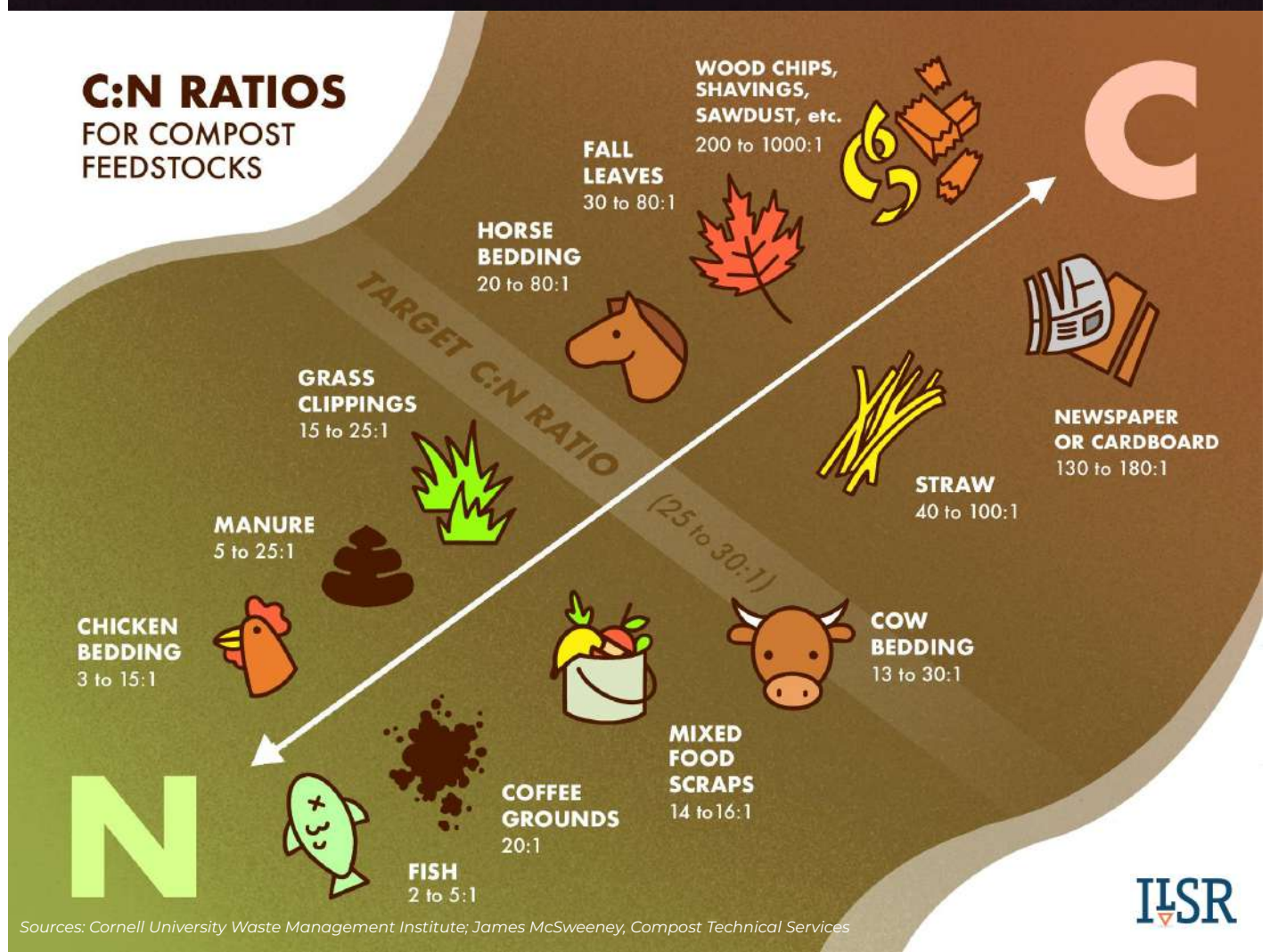


Carbon to Nitrogen Ratio (C:N)

Materials High in Nitrogen	C:N Ratio
Manure	5 - 25:1
Grass clippings	15 - 25:1
Coffee grounds	20:1
Vegetable wastes	15 - 20:1

Materials High in Carbon	C:N Ratio
Newspaper or cardboard	130 - 180:1
Bark	100 - 130:1
Wood chips or sawdust	200 - 1,000:1
Fall leaves	30 - 80:1

Source: Cornell University Waste Management Institute



Sources: Cornell University Waste Management Institute; James McSweeney, Compost Technical Services



Carbon to Nitrogen Ratio (C:N)

Recipe calculator

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Recipe Results **RESET**
3 feedstocks

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GENERAL GUIDELINE

2-3 parts **BROWNS** to
1 part **GREENS** (by volume)



GREENS
(nitrogen-rich material)



BROWNS
(carbon-rich material)



Moisture Content

Moisture should be between 50-60%

- If too dry, microbial activity will slow or cease
- If too wet, may lead to anaerobic conditions



Benny Erez at ECO City Farms demonstrating how to ensure adequate water is mixed well into recipe before adding to compost bin.

Photo: ILSR



THE "HAND SQUEEZE" MOISTURE TEST



Take handfuls from different places in your compost pile.

Give them a squeeze!

At the ideal moisture level, a few drops of water should appear between your knuckles like a wrung-out sponge.



Air & Bulk Density

Bulk density

= an estimation of porosity
= weight of a certain
volume of a material

Starting
optimal range:
800 and 1,000
pounds
per cubic
yard



Red Hook Community Farm Compost Operation, Brooklyn, NYC.

Photos: ILSR

Pay Attention to Density!

TOO LIGHT

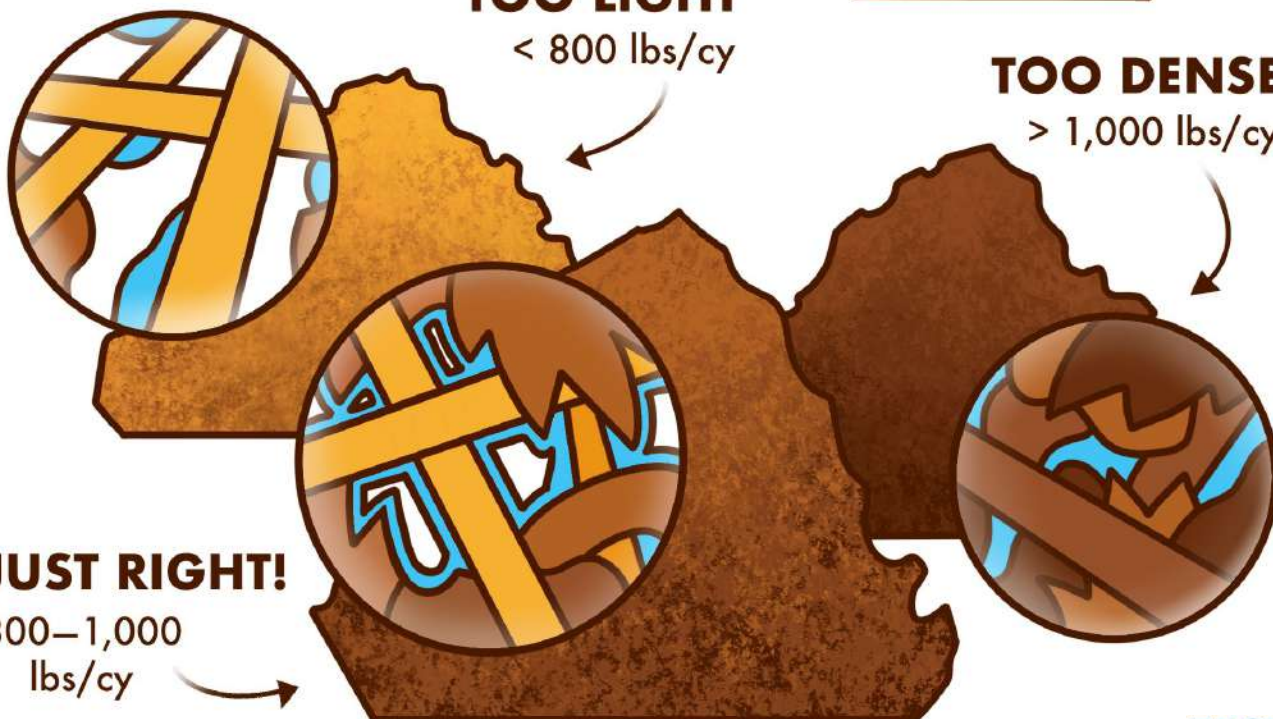
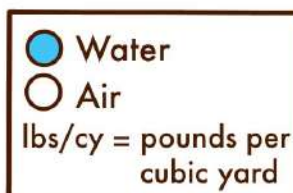
< 800 lbs/cy

TOO DENSE

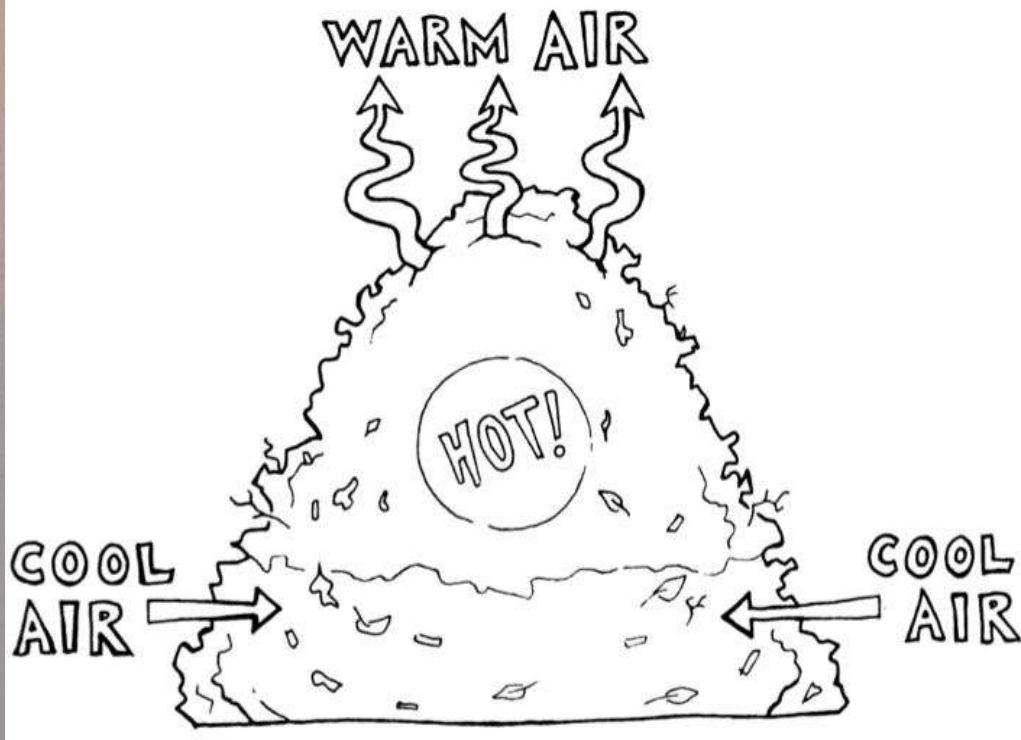
> 1,000 lbs/cy

JUST RIGHT!

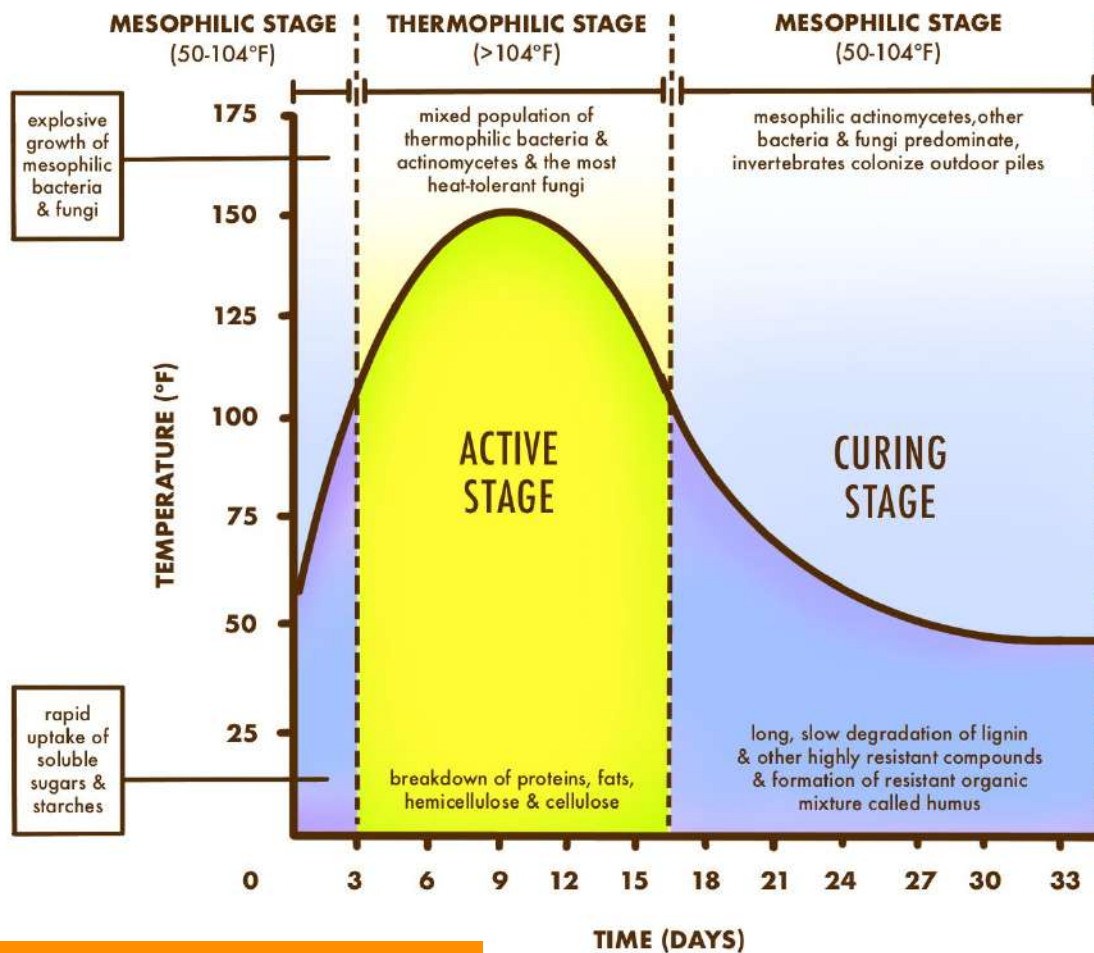
800–1,000
lbs/cy



The “Chimney Effect”



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Different Phases, Different Microbes

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Composting Fundamentals & Recipe Development

- ✓ Compost Pile Food Web
- ✓ Types of Composting
- ✓ Ingredients & Factors
- ✓ Composting Recipes
- ✓ Temperature & Composting Stages



Photo: @LACompost



Building a Compost Pile

- ✓ Recommendations for Materials to Accept & Avoid
- ✓ The Right Mix of Browns
- ✓ Measuring Moisture Content & Bulk Density
- ✓ Preparing Materials for Composting
- ✓ Methods for Building a Pile

